CH06

- $x[n]$이 FIR filter를 거쳐서 $y[n]$

  same frequency

  different (Amplitude, Phase) → frequency response $H(e^{j\hat{\omega}})$

- $x[n] = Ae^{j\varphi} e^{j\hat{\omega}n}$

$$y[n] = \sum_{k=0}^{M} h[k]\, x[n-k]$$

$$= \sum_{k=0}^{M} h[k]\, Ae^{j\varphi} e^{j\hat{\omega}(n-k)}$$

$$= Ae^{j\varphi} e^{j\hat{\omega}n} \left(\sum_{k=0}^{M} h[k] e^{-j\hat{\omega}k} \right)$$

$\left(\sum_{k=0}^{M} h[k] e^{-j\hat{\omega}k} \right) = H(e^{j\hat{\omega}})$: frequency response

$$\therefore y[n] = H(e^{j\hat{\omega}})\, x[n]$$

$$= \left(\sum_{k=0}^{M} b_k e^{-j\hat{\omega}k} \right) x[n]$$

- $H(e^{j\hat{\omega}}) = |H(e^{j\hat{\omega}})| \cdot e^{j\angle H(e^{j\hat{\omega}})}$

  magnitude, phase

  (Mag는 $\hat{\omega}$에 대한 짝함수, phase는 홀함수)

- 예제 $b_k = \{1, 2, 1\}$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} b_k e^{-j\hat{\omega}k} = 1 \times e^{-j\hat{\omega}\times 0} + 2 \times e^{-j\hat{\omega}\times 1} + 1 \times e^{-j\hat{\omega}\times 2}$$

$$= 1 + 2e^{-j\hat{\omega}} + e^{-j2\hat{\omega}}$$

$$= e^{-j\hat{\omega}} \left(e^{j\hat{\omega}} + 2 + e^{-j\hat{\omega}} \right) = e^{-j\hat{\omega}} \underbrace{(1 + 2\cos\hat{\omega})}_{\text{magnitude}}$$

$\underbrace{\quad}_{\text{phase} - \hat{\omega}}$

- Euler's formula

$$\cos(w) = \frac{1}{2} e^{jw} + \frac{1}{2} e^{-jw}$$

$$\sin(w) = \frac{1}{2j} e^{jw} - \frac{1}{2j} e^{-jw}$$

- 예제

$$x[n] = 2e^{j\frac{\pi}{4}} e^{j\frac{\pi}{3}n} \qquad \text{frequency} : \frac{\pi}{3}$$

$$H(e^{j\hat{\omega}}) = (2 + 2\cos\hat{\omega})\, e^{-j\hat{\omega}} \qquad \leftarrow H(e^{j\hat{\omega}}) \text{에 대입}$$

$$= (2 + 2\cos\tfrac{\pi}{3})\, e^{-j\frac{\pi}{3}}$$

$\cos\frac{\pi}{3} = \frac{1}{2}$

$$= (2+1)\, e^{-j\frac{\pi}{3}} = 3e^{-j\frac{\pi}{3}}$$

$$y[n] = H(e^{j\hat{\omega}}) \cdot x[n] = 3e^{-j\frac{\pi}{3}} \left(2e^{j\frac{\pi}{4}} e^{j\frac{\pi}{3}n} \right)$$

$$= 6\, e^{-j\frac{1}{12}\pi} e^{j\frac{\pi}{3}n} \rightarrow \begin{cases} \text{magnitude} : 6 \\ \text{phase} : -\frac{\pi}{12} \end{cases}$$

- 예제

$$2\cos\left(\frac{\pi}{3}n + \frac{\pi}{4}\right) = 2\left(\frac{1}{2} e^{j\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)} + \frac{1}{2} e^{-j\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)} \right)$$

$$= \underbrace{e^{j\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)}}_{①} + \underbrace{e^{-j\left(\frac{\pi}{3}n + \frac{\pi}{4}\right)}}_{②}$$

① $y_1[n] = H(e^{j\frac{\pi}{3}})\, e^{j(\frac{\pi}{3}n+\frac{\pi}{4})}$ $\qquad (2+2\cos\hat{\omega})e^{-j\hat{\omega}}$

$\downarrow$

$3e^{j\frac{\pi}{3}}\, e^{j\frac{\pi}{3}n+\frac{\pi}{4}}$

$= 3e^{j(\frac{\pi}{4}-\frac{\pi}{3})}\, e^{j\frac{\pi}{3}n} = 3e^{-j\frac{\pi}{12}}\, e^{j\frac{\pi}{3}n}$

② $y_2[n] = H(e^{j(-\frac{\pi}{3})})\, e^{-j(\frac{\pi}{3}n+\frac{\pi}{4})}$ $\qquad = 3e^{j\frac{\pi}{3}}\, e^{-j(\frac{\pi}{3}n+\frac{\pi}{4})}$

$\downarrow$

$(2+2\cos(-\frac{\pi}{3}))\, e^{+j\frac{\pi}{3}}$ $\qquad = 3e^{j\frac{\pi}{12}}\, e^{-j\frac{\pi}{3}n}$

$\downarrow \frac{1}{2}$

$3e^{j\frac{\pi}{3}}$

$3e^{-j\frac{\pi}{12}}\, e^{j\frac{\pi}{3}n} + 3e^{j\frac{\pi}{12}}\, e^{-j\frac{\pi}{3}n}$

$= 3e^{j(\frac{\pi}{3}n-\frac{\pi}{12})} + 3e^{-j(\frac{\pi}{3}n-\frac{\pi}{12})} = 6\cos(\frac{\pi}{3}n-\frac{\pi}{12})$

- $\sum_{k=0}^{M} h[k]\,\delta[n-k] \longleftrightarrow \sum_{k=0}^{M} h[k]e^{-j\hat{\omega}k}$

- 예제 $y[n]=x[n-1]$ 일 때.

$h[n]=\delta[n-1]$ $\qquad \therefore b_k=\{0,1\}$

$H(e^{j\hat{\omega}}) = \sum_{k=0}^{M} h[n]\, e^{-j\hat{\omega}k} = \cancel{h[0]}\cdot e^{0} + h[1]\cdot e^{-j\hat{\omega}}$

$= e^{-j\hat{\omega}}$

- 예제1 $y[n] = x[n] - x[n-1]$ 일 때

$h[n] = \delta[n] - \delta[n-1]$ $\quad b_k = \{1, -1\}$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{1} h[n]\, e^{-j\hat{\omega}k} = 1 \times e^{-j\hat{\omega}\cdot 0} - 1 \times e^{-j\hat{\omega}\cdot 1}$$
$$= \underline{1 - e^{-j\hat{\omega}}}$$

- 예제2 $y[n] = x[n-2]$ 일 때

$h[n] = \delta[n-2]$ $\quad b_k = \{0, 0, 1\}$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{2} h[n]\, e^{-j\hat{\omega}k} = 0 \cdot e^{-j\hat{\omega}\cdot 0} + 0 \cdot - + 1 \cdot e^{-j\hat{\omega}\cdot 2}$$
$$= e^{-j2\hat{\omega}}$$

- general delay

$y[n] = x[n - n_d]$ 이면 $\quad h[n] = \delta[n - n_d]$

$$\therefore H(e^{j\hat{\omega}}) = \sum_{k=0}^{m} \delta[n - n_d]\, e^{-j\hat{\omega}k} = \underline{e^{-j\hat{\omega} n_d}}$$

- $H(e^{j\hat{\omega}}) = 7e^{-j2\hat{\omega}}\cos(\hat{\omega})$ 일 때 $h[n]$?

↓

$7e^{-j2\hat{\omega}}\left(\frac{1}{2}e^{j\hat{\omega}} + \frac{1}{2}e^{-j\hat{\omega}}\right)$ $\quad \delta[n - n_d] \rightleftarrows e^{-j\hat{\omega} n_d}$

$= 3.5\underbrace{e^{-j\hat{\omega}}}_{\delta[n-1]} + 3.5\underbrace{e^{-j3\hat{\omega}}}_{\delta[n-3]}$

$h[n] = 3.5\,\delta[n-1] + 3.5\,\delta[n-3]$ $\quad \therefore b_k = \{0, 3.5, 0, 3.5\}$

Cascade LTI, Equivalent system

$x[n] \rightarrow H_1(e^{j\hat{\omega}}) \rightarrow H_2(e^{j\hat{\omega}}) \rightarrow y[n]$

$= H_1(e^{j\hat{\omega}}) \times H_2(e^{j\hat{\omega}})$

$h_1[n] * h_2[n]$

$=$

$h_2[n] * h_1[n]$

Commutative property

- 11 point Avg

$$y[n] = \sum_{k=0}^{10} \frac{1}{11} x[n-k] \qquad b_k = \frac{1}{11} \qquad k = 0, \cdots, 10$$

$$H(e^{j\hat{\omega}}) = \sum_{k=0}^{10} b_k e^{-j\hat{\omega}k} = \sum_{k=0}^{10} \frac{1}{11} e^{-j\hat{\omega}k} = \frac{1}{11} \sum_{k=0}^{10} e^{-j\hat{\omega}k}$$

same as $e^{j\hat{\omega}}$

$$= \frac{1}{11} \frac{1-(e^{-j\hat{\omega}})^{11}}{1-e^{-j\hat{\omega}}} = \frac{1}{11} \cdot \frac{e^{-j\frac{11}{2}\hat{\omega}}\left(e^{j\frac{11}{2}\hat{\omega}} - e^{-j\frac{11}{2}\hat{\omega}}\right)}{e^{-j\frac{1}{2}\hat{\omega}}\left(e^{j\frac{1}{2}\hat{\omega}} - e^{-j\frac{1}{2}\hat{\omega}}\right)}$$

$$= \frac{1}{11} e^{-j5\hat{\omega}} \frac{2j\sin\left(\frac{11}{2}\hat{\omega}\right)}{2j\sin\left(\frac{1}{2}\hat{\omega}\right)} = \frac{\sin\left(\frac{11}{2}\hat{\omega}\right)}{11\sin\left(\frac{1}{2}\hat{\omega}\right)} e^{-j5\hat{\omega}}$$

- filter types
  - Low pass filter
  - high pass filter
  - band pass filter